

Supertranslations in Ashtekar-Barbero variables

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"Boundary conditions for Ashtekar-Barbero variables in the context of asymptotically flat spacetimes which lead to supertranslations at spatial infinity," **Sepideh Bakhoda Nov. 2, 2023**

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Overview

- **Regge and Teitelboim** (1974) provided boundary conditions known as **RT** or **standard** boundary conditions at spatial infinity, which yielded charges of the **poincare** group (energy, momentum, angular momentum, and boost) as required surface integrals of the Hamiltonian.
- **Thiemann** (1995) showed the classical equivalence of the metric (ADM) variables and Ashtekar (triad-type) variables for **RT** boundary conditions, (i.e. a canonical transformation) when $\beta = i$. **Campiglia** (2014) showed the classical equivalence for all real values of β .
- **Henneaux and Troessaert** (2018) developed new **HT** boundary conditions, consistent with the asymptotic structure, which yielding non vanishing **poincare** and **supertranslation** charge
- **Bakhoda** (2023) showed for the **HT** boundary conditions; there does **not** exist a canonical transformation between the ADM and Ashtekar variables. The nonvanishing charges in this scenario are energy, momentum, and super translation.

For any gauge theory, the Hamiltonian terms are implemented as local **constraints** on phase space and vanish on the **physical constraint surface**. In the ADM formulation of gravity, the three metric on a spacelike hypersurface and its conjugate momenta $(q_{ab}(x), \pi^{ab}(x))$ are constrained to evolve along t and x_i by **Lapse**: $\mathcal{N}$ and **shift**: $\mathcal{N}_i$ respectively. For **compact** spacetimes:

$$H(\mathcal{N}, \mathcal{N}_i) = \int d^3x \{ \mathcal{N} \mathcal{H}(x) + \mathcal{N}_i \mathcal{H}^i(x) \}$$

The symplectic structure is well defined if its of the following form and **converges**:

$$\delta H = \int d^3x \left(\delta \mathcal{N} q_{ab} \delta \pi^{ab} - \delta \mathcal{N} \pi^{ab} \delta q_{ab} \right) \quad (\delta \mathcal{N} q_{ab} = \delta_{\mathcal{N}, \mathcal{N}_i} q_{ab} = \delta H / \delta q_{ab})$$

For **non compact** spacetimes, the asymptotic "fall-off" behavior, is defined outside a large compact region and δH may contain total derivatives thus surface terms, an apparent spoiling of well-definiteness unless it is also **exact** as follows:

$$\begin{aligned} \delta H &= \delta \int d^3x \left(\mathcal{N} \mathcal{H}(x) + \mathcal{N}_i \mathcal{H}^i(x) \right) + \lim_{r \rightarrow \infty} \int d^2x \mathcal{K}_{\mathcal{N}}(\delta q_{ab}, \delta \pi^{ab}) \\ \mathcal{K}_{\mathcal{N}}(\delta q_{ab}, \delta \pi^{ab}) &= -\delta \tilde{\mathcal{B}}_{\mathcal{N}}(q_{ab}, \pi^{ab}) \\ \delta(H + \mathcal{B}_{\mathcal{N}}) &= \delta \int d^3x \left(\mathcal{N} \mathcal{H}(x) + \mathcal{N}_i \mathcal{H}^i(x) \right) \end{aligned}$$

Asymptotically flat spacetimes limit to a flat 3-metric $q_{ab} = \delta_{ab}$, $h_{ab}(\hat{x}) = h_{ab}(-\hat{x})$

$$q_{ab} = \delta_{ab} + \frac{\bar{h}_{ab}}{r} + \frac{h_{ab}^{(2)}}{r^2} + o(r^{-2}) \quad \pi^{ab} = \frac{\bar{\pi}^{ab}}{r^2} + \frac{\pi_{ab}^{(2)}}{r^3} + o(r^{-3})$$

B.C. are dynamically preserved so $\pi^{ab} \sim \text{odd}$ for finite $\int d^3x \delta_{\mathcal{N}} q_{ab} \delta \pi^{ab}$

In spherical coordinates:

$$q_{rr} = 1 + \frac{\bar{h}_{rr}}{r} + \frac{h_{rr}^{(2)}}{r^2} + o(r^{-2}) \quad \pi^{rr} = \bar{\pi}^{rr} + \frac{\pi^{(2)rr}}{r} + o(r^{-1})$$

$$q_{rA} = \frac{h_{rA}^{(2)}}{r} + o(r^{-1}) \quad \pi^{rA} = \frac{\bar{\pi}^{rA}}{r^2} + \frac{\pi^{(2)rA}}{r^2} + o(r^{-2})$$

$$q_{AB} = r^2 \bar{\gamma}_{AB} + r \bar{h}_{AB} + h_{AB}^{(2)} + o(1) \quad \pi^{AB} = \frac{\bar{\pi}^{AB}}{r^2} + \frac{\pi^{(2)AB}}{r^2} + o(r^{-3})$$

Regge-Teitelbom standard b.c.	Henneaux -Troessaert mixed b.c.
$\bar{h}_{rr} \sim \bar{h}_{\theta\theta} \sim \bar{h}_{\phi\phi} \sim \bar{\pi}^{r\theta} \sim \bar{\pi}^{\theta\phi} = \text{even}$	$\bar{\lambda} \sim \bar{\pi}^{r\phi} \sim \bar{\pi}^{\theta\theta} \sim \bar{\pi}^{\phi\phi} \sim \bar{k}_{\theta\phi} = \text{even}$
$\bar{\pi}^{rr} \sim \bar{\pi}^{r\theta} \sim \bar{\pi}^{\phi\phi} \sim \bar{\pi}^{\theta\theta} \sim \bar{h}_{\theta\phi} = \text{odd}$	$\bar{p} \sim \bar{\pi}^{r\theta} \sim \bar{\pi}^{\theta\phi} \sim \bar{k}_{\theta\theta} \sim \bar{k}_{\phi\phi} = \text{odd}$

Where $\bar{p} = 2(\bar{\pi}^{rr} - \bar{\pi}_A^A)$, $\bar{\lambda} = \frac{\bar{h}_{rr}}{2}$, $\bar{k}_{AB} = \frac{\bar{h}_{AB}}{2} + \bar{\lambda} \bar{\gamma}_{AB}$

The Lapse and shift in spherical coordinates is of the form:

$$N = rb + f + O(r^{-1}) \quad N^r = W + O(r^{-1})$$

$$N^A = Y^A + \frac{1}{r} \left(\frac{2b}{\sqrt{\gamma}} \pi^{rA} + \bar{D}^A W \right) + o(r^{-2})$$

For **boost**: $b = b_1 \sin \theta \cos \phi + b_2 \sin \theta \sin \phi + b_3 \cos \theta$. For **rotations**, Y^A are the standard killing vectors on the sphere. W/f are **spatial/time (super) translations**. Usual translations are: $W_P = \sum_{m=-1}^1 P^m Y_m^1, f = f_0$.

(**RT**) : $f - f_0 = \text{odd}, W - W_P = \text{even}$ (**HT**) : $f = \alpha + T, T = \text{even}, W = \text{odd}$

The divergence in the charge given by:

$$\mathcal{K}_{\mathbf{N}}^{\text{div}}(\delta q_{ab}, \delta \pi^{ab}) = r \int d^2 x \{ 2Y^A \bar{\gamma}_{AB} \bar{D}_C \delta \bar{\pi}^{BC} - 2\sqrt{\gamma} b \delta(\bar{k} - \bar{D}_A \bar{D}_B \bar{k}^{AB} + \bar{D}_A \bar{D}^A \bar{k}) \}$$

is removed by setting $\bar{H}_A = \bar{H} = 0$ in **HT** and by parity in **RT**. The charges:

$$\mathcal{B}_{\mathbf{N}}(q_{ab}, \pi^{ab}) = \int d^2 x \{ T 4\sqrt{\gamma} \bar{\lambda} + W \bar{p} + Y^A 2\gamma_{AB} (\pi^{(2)rB} - 2\bar{\lambda} \bar{\pi}^{rB})$$

$$+ b 2\sqrt{\gamma} (k^2 - 3\bar{\lambda} \bar{k}) \}$$

For non-constant T/W (supertranslations), **RT** parity leads to a vanishing integral, **HT** parity gives finite super translation charge.

Instead of ADM phase space, we consider the **Ashtekar-Barbero (AB)** phase space of (densitized) triads

$$\mathbf{E}_i^a(x) = \sqrt{\det(q_{ab})} e_i^a, \text{ where } q^{ab} = e_i^a e_j^b \delta_{ij}$$

and it's conjugate momenta (immerzi parameter- introduce the holts action)

$$\mathbf{A}_a^i(x) = \Gamma_a^i + \beta K_a^i \quad K_{ab} = K_{[a}^j e_{b]}^j \quad \Gamma_a^i = \epsilon_{ijk} \Gamma_{ka}^j$$

which are SU(2) representations labeled by: $i=x,y,z$. For their to exist a **canonical transformation** between (**AB**) variables and ADM, the generator of internal SU(2) rotations, $\mathcal{G}_i(\Lambda^i)$, must be a constraint and vanish on-shell

$$\mathcal{G}_i(\Lambda^i) = \frac{2}{\beta} \int_{\Sigma} d^3x \Lambda^i (\partial_a E_i^a + \epsilon_{ijk} A_a^j E_k^a) = \frac{2}{\beta} \int_{\Sigma} d^3x \Lambda^i \mathcal{D}_a E_i^a$$

In addition the poisson structure must be the same, to ensure $\{A_a^j, A_b^k\} = 0$ the following must be true: $\frac{\delta \Gamma_a^j(x)}{\delta E_k^b(y)} = \frac{\delta \Gamma_b^k(x)}{\delta E_j^a(y)}$ which is easily satisfied for boundary-less manifolds, and the standard parity conditions, but fails to hold for the **HT** boundary conditions

The expansion and boundary conditions for the **AB** variables are as follows:

$$E_i^r = r^2 \sqrt{\bar{\gamma}} \bar{\gamma}_i^r + r \sqrt{\bar{\gamma}} \bar{f}_i^r + \sqrt{\bar{\gamma}} \bar{f}_i^{r(2)} + O(r^{-1}) \quad A_r^i = \frac{\bar{g}_r^i}{r^2} + \frac{\bar{g}_r^{(2)i}}{r^3} + o(r^{-4})$$

$$E_i^A = r \sqrt{\bar{\gamma}} \bar{\gamma}_i^A + \sqrt{\bar{\gamma}} \bar{f}_i^A + \sqrt{\bar{\gamma}} \frac{\bar{f}_i^{r(2)}}{r} + O(r^{-1}) \quad A_B^i = \frac{\bar{g}_B^i}{r} + \frac{\bar{g}_B^{(2)i}}{r^2} + o(r^{-3})$$

Fixing $\bar{\gamma}_a^i = \delta_a^i$ and $\delta \bar{\gamma}_a^i = 0$, consider the quantities: $\bar{F}_b^a = \bar{f}_i^a \bar{\gamma}_b^i$, and $\bar{G}_b^a = \bar{g}_b^i \bar{\gamma}_i^a$
 $\bar{\gamma}_i^r \sim \bar{\gamma}_i^\phi = \text{odd}$, $\bar{\gamma}_i^\theta = \text{even}$

$$\bar{F}_r^\theta \sim \bar{F}_\phi^\theta \sim \bar{F}_\theta^\phi \sim (\bar{F}_r^r - \bar{F}_A^A) \sim (\bar{G}_\theta^\theta + \bar{G}_r^r) = \text{even}$$

$$\bar{G}_\phi^r - \frac{\sqrt{\bar{\gamma}} \bar{D}_\theta (\bar{F}_r^r - \bar{F}_A^A)}{2} \sim \bar{G}_r^\phi - \frac{\bar{D}_\theta (\bar{F}_r^r - \bar{F}_A^A)}{2\sqrt{\bar{\gamma}}} = \text{even}$$

$$\bar{G}_\theta^r + \frac{\bar{D}_\theta (\bar{F}_r^r - \bar{F}_A^A)}{2\sqrt{\bar{\gamma}}} \sim \bar{G}_r^\theta + \frac{\bar{D}_\theta (\bar{F}_r^r - \bar{F}_A^A)}{2\sqrt{\bar{\gamma}}} \sim \bar{G}_\theta^\phi - \frac{\bar{D}_\theta (\bar{F}_r^r - \bar{F}_A^A)}{2\sqrt{\bar{\gamma}}} = \text{odd}$$

$$\bar{G}_\phi^\theta + \frac{\sqrt{\bar{\gamma}} \bar{D}_\theta (\bar{F}_r^r - \bar{F}_A^A)}{2} \sim \bar{F}_r^\phi \sim \bar{F}_\theta^\theta \sim \bar{F}_\phi^\phi \sim \bar{G}_r^r = \text{odd}$$

The asymptotic expansion of $\Lambda^i = \frac{\bar{\Lambda}^i}{r} + O(r^{-1})$ with odd $\bar{\Lambda}^i$ in **RT** and even $\bar{\Lambda}^i$ in **HT**. The SU(2) charge vanishes for **RT** but not for **HT**!

$$\delta \mathcal{G}_i(\Lambda^i) = \int d^3x (\delta_{\Lambda} A_a^k \delta E_k^a - \delta_{\Lambda} E_k^a \delta A_a^k) + \frac{2}{\beta} \int dS^a (\bar{\Lambda}^i \bar{f}_a^i)$$

To ensure $\delta \bar{\gamma}_a^i = 0$ we must modify the generators

$$\mathcal{H}_{\mathbf{AB}}^{(asym)}(\mathbf{N}) = \mathcal{H}_{\mathbf{AB}}(\mathbf{N}) - \frac{1}{8\pi G} \int dS_a \mathbf{N} \epsilon_{ijk} A_b^i E_j^a E_k^b - \beta \mathcal{G}(\Lambda_B) + \frac{1}{8\pi G} \int dS_a E_a^i \bar{\Lambda}_B^i \quad (1)$$

$$\mathcal{H}_{\mathbf{AB}}^{i(asym)}(N_i) = \mathcal{H}_{\mathbf{AB}}^i(N_i) - \frac{1}{8\pi G \beta} \int dS_a N^r A_b^i E_i^b - \mathcal{G}(\Lambda_R^i) + \frac{1}{8\pi G \beta} \int dS_a E_a^i \bar{\Lambda}_R^i \quad (2)$$

The $\bar{\Lambda}_R^i = \epsilon_{ijk} Y^B \bar{\gamma}_B^k \bar{\gamma}_j^r$, $\bar{\Lambda}_B^i = 2b \bar{\gamma}_i^r$.

$$\begin{aligned} \delta_{(\mathbf{N}, \Lambda^i)} \int d^3x (\mathcal{H}_{\mathbf{AB}}(\mathbf{N}) + \mathcal{H}_{\mathbf{AB}}^i(N_i) + \mathcal{G}(\Lambda^i)) \\ = \frac{2}{\beta} \int d^3x (\delta_{(\mathbf{N}, \Lambda^i)} E_a^i \delta A_a^i - \delta_{(\mathbf{N}, \Lambda^i)} A_a^i \delta E_a^i) + \mathcal{B}_{(\mathbf{N}, \Lambda^i)}(\delta A_a^i, \delta E_a^i) \end{aligned}$$

The charge $\mathcal{B}_{(N,\Lambda^i)}(\delta A_i^a, \delta E_a^i)$ has divergent terms expressed as:

$$\begin{aligned}\mathcal{B}_{(N,\Lambda^i)}^{\text{div}}(\delta A_i^a, \delta E_a^i) = & 2r \int d\Omega \mathbf{b} \epsilon^{AB} \bar{\gamma}_{BC} \delta \bar{G}_A^C - \mathbf{b} \sqrt{\bar{\gamma}} \delta \bar{F}_r^r - \sqrt{\bar{\gamma}} (\partial_A \mathbf{b}) \bar{\gamma}^{AB} \delta \bar{F}_B^r \\ & + \frac{-1}{\beta} \sqrt{\bar{\gamma}} \mathbf{Y}^A (\delta \bar{G}_A^r - 2\epsilon^{BC} \bar{\gamma}_{AC} \delta \bar{F}_B^r)\end{aligned}$$

However, the conditions $\bar{\mathcal{H}} = \bar{\mathcal{H}}^i = \bar{\mathcal{G}} = 0$ do not cause $\mathcal{B}_{(N,\Lambda^i)}^{\text{div}}$ to vanish!

To get well defined dynamics we need to set $Y^A = b = 0$

$$\mathcal{B}_{(N,\Lambda^i)}(\delta A_i^a, \delta E_a^i)|_{Y^A=b=0} = \delta 2 \int d\Omega \sqrt{\bar{\gamma}} (\bar{f}(\bar{F}_r^r - \bar{F}_A^A) - \frac{-2}{\beta} W \bar{G}_r^r) = \delta \mathcal{Q}_{\text{supertrans}}$$

where constant f, W are usual translations associated to energy and momentum.

In summary:

	ADM	AB
RT	energy, momentum, angular momentum, boost	energy, momentum, angular momentum, boost
HT	energy, momentum, angular momentum, boost, supertranslation	energy, momentum, supertranslation

Goals

- Find boundary conditions in Ashtekar-Barbero variables for which the asymptotic dynamics are well defined and the BMS charges are non-vanishing
- Compute poisson brackets between the charges and implement them as phase space constraints
- Promote to quantum operators and determine effect on eigenstates of canonical variables (spin network functions)

Citations

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